AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2013

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

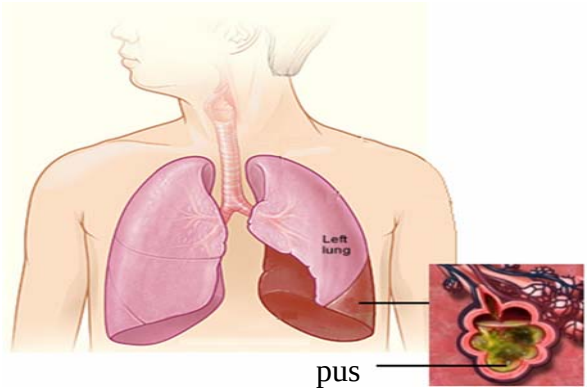
3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

The diagram shows the lungs of a patient suffering from one of the respiratory diseases.



a. Identify the disease and state its cause. (2 Marks)

_____

_____

_____

_____

b. State any TWO symptoms that this patient might be displaying. (2 Marks)

_____

_____

_____

_____

Q.2. (Total 4 Marks)

a. What happens to carbon dioxide in plants in daytime and at night? (2 Marks)

In day time: _____

_____

At night: _____

_____

b. Draw a ray diagram to show how long sightedness can be corrected using a lens. (2 Marks)

Space for diagram

Q.3. (Total 4 Marks)

What is antagonism? Why are biceps and triceps considered as the antagonistic muscles?

Q.4. (Total 4 Marks)

a. Why is stem cutting considered as an artificial and vegetative method of propagation in plants? (2 Marks)

_____

_____

_____

_____

b. Why is the flow of energy considered as non-cyclic in an ecosystem? (2 Marks)

_____

_____

_____

_____

Q.5. (Total 4 Marks)

The given picture shows the habitat of butterflies of two different colours. One variety of butterflies appears more in number than the other.



Describe the phenomenon responsible for this discriminating survival.

Q.6. (Total 5 Marks)

a. State any TWO objectives behind the implementation of genetically engineered organisms. (2 Marks)

_____

_____

_____

_____

b. What physiological changes occur in the body of a person taking hallucinogens? Mention any THREE. (3 Marks)

_____

_____

_____

_____

_____

_____

Q.7. (Total 7 Marks)

EITHER

a. Explain the process of the two types of dialysis carried out in case of kidney failure.

OR

b. Think of an example of a reflex action from your daily life. Describe the events/changes taking place from the time you sense it till the point you respond to it.

(Your answer should also include the complete pathway of nerve impulses.)

Q.8. (Total 8 Marks)

EITHER

a. Describe the structural features which are found in a wind pollinated and an insect pollinated flower.

OR

b. Describe how urbanization and acid rain cause harm to the ecosystem.

END OF PAPER

Please use this page for rough work